

DIGITAL FORENSIC EXAMINATION REPORT

UNMANNED AERIAL VEHICLE (UAV) FLIGHT DATA & TELEMETRY EXTRACTION

1. INVESTIGATIVE CASE ADMINISTRATION

Case Reference:	CASE-VERIFY-001	Evidence Item ID:	real_px4_flight.ulg
Examining Agency:	Cyber Forensic Investigation Laboratory (CFSL / State Police)	Lead Examiner:	Forensic Examiner
Examination Date:	2026-09-05 17:14:58 UTC	Forensic Engine:	Pushpak Indigenous Drone Forensics Toolkit (v1.0.0-phase1 (Build 2026.09))

2. EVIDENCE INTEGRITY & CHAIN OF CUSTODY

Source Evidence File:	C:\Users\pawan\Desktop\df\sample_evidence\real_px4_flight.ulg
File Size:	27,388,166 bytes
SHA-256 Digest:	b8abd99f208de7fd072d2e271524507b761e061e6cda9170685caf85901ca027
BLAKE3 Digest:	f6f26e52465feb6e27339cf256dde2a6469252f44085e80616e1709ac3ff46dc
Chain Verification:	VERIFIED INTACT (100% Cryptographic Match)

3. FLIGHT MISSION ENVELOPE & TELEMETRY PROFILE

Platform / Architecture:	PX4	Aircraft Model:	PX4_FMU_V6X
Hardware Serial No:	N/A	Total Telemetry Records:	1,928 events
Takeoff Time (UTC):	2026-09-05 05:42:27 UTC	Termination Time (UTC):	2026-09-05 05:45:19 UTC
Total Flight Duration:	171.6 seconds (2.9 mins)	Total GPS Waypoints:	859 points
Max Altitude (AGL):	113.0 meters (370.8 ft)	Peak Ground Velocity:	29.0 m/s (104.4 km/h)

4. FORENSIC ANOMALY FINDINGS & THREAT ASSESSMENT

Anomaly Type	Severity	Timestamp (UTC)	Location / Alt	Technical Finding Narrative
IMPACT OR CRASH	CRITICAL	05:45:09 UTC	N/A	Catastrophic attitude deviation / impact shock: pitch=38.3°, roll=-97.0°
IMPACT OR CRASH	CRITICAL	05:45:09 UTC	35.1913, 128.1476 Alt: 60.4m	Catastrophic attitude deviation / impact shock: pitch=1.2°, roll=-142.8°
IMPACT OR CRASH	CRITICAL	05:45:09 UTC	N/A	Catastrophic attitude deviation / impact shock: pitch=15.5°, roll=-131.1°
IMPACT OR CRASH	CRITICAL	05:45:10 UTC	35.1913, 128.1476 Alt: 59.2m	Catastrophic attitude deviation / impact shock: pitch=1.2°, roll=-142.8°
IMPACT OR CRASH	CRITICAL	05:45:10 UTC	N/A	Catastrophic attitude deviation / impact shock: pitch=-16.2°, roll=-154.8°

IMPACT OR CRASH	CRITICAL	05:45:10 UTC	N/A	Catastrophic attitude deviation / impact shock: pitch=-48.5°, roll=177.9°
IMPACT OR CRASH	CRITICAL	05:45:10 UTC	35.1913, 128.1476 Alt: 57.5m	Catastrophic attitude deviation / impact shock: pitch=-62.9°, roll=91.0°
IMPACT OR CRASH	CRITICAL	05:45:10 UTC	N/A	Catastrophic attitude deviation / impact shock: pitch=-64.8°, roll=119.5°
IMPACT OR CRASH	CRITICAL	05:45:10 UTC	35.1913, 128.1476 Alt: 55.5m	Catastrophic attitude deviation / impact shock: pitch=-62.9°, roll=91.0°
IMPACT OR CRASH	CRITICAL	05:45:10 UTC	35.1913, 128.1476 Alt: 53.1m	Catastrophic attitude deviation / impact shock: pitch=-62.9°, roll=91.0°
IMPACT OR CRASH	CRITICAL	05:45:14 UTC	N/A	Catastrophic attitude deviation / impact shock: pitch=46.1°, roll=-88.1°
IMPACT OR CRASH	CRITICAL	05:45:14 UTC	35.1912, 128.1476 Alt: 37.2m	Catastrophic attitude deviation / impact shock: pitch=14.0°, roll=-148.3°

5. FORENSIC CONFIDENCE & UNCERTAINTY ASSESSMENT

Overall Forensic Confidence Rating: DEGRADED (POOR SATELLITE COVERAGE) - 75.0%
In accordance with ISO/IEC 27037:2012 guidelines, this assessment explicitly defines the mathematical certainty and evidentiary boundaries of the reconstructed flight profile.

Forensic Dimension	Confidence Level	Evidentiary Basis & Limitations
Bit-Stream Integrity	ABSOLUTE (100%)	Dual SHA-256 + BLAKE3 hashes match original media bit-stream.
Spatial Trajectory	HIGH (75.0%)	Avg Satellites: 0.0 Avg HDOP: 0.24 Multipath risk low.
Temporal Synchronization	HIGH (99.0%)	Correlated with GPS atomic UTC reference clock.
Evidence Limitations	DOCUMENTED	Barometric altitude uncorrected for local QNH pressure.

6. MARKED 3D TRAJECTORY FORENSIC EXHIBITS

High-resolution photographic captures of 3D aerospace trajectory reconstruction with itemized incident kinematics, spatial position, and forensic findings recorded at marked waypoints.

EVID-01: 3D Trajectory Incident Reconstruction (Waypoint #11 at 12:44:19 UTC)



Waypoint Ref:	Waypoint #11	Incident Time UTC:	12:44:19 UTC
Altitude (MSL):	28.5 m	Ground Speed:	18.2 m/s
Flight Heading:	90° (E)	Attitude (Pitch/Roll):	Pitch: -12.4° Roll: +8.5°
GNSS Position:	19.076012°, 72.877715°	Custody Verification:	VERIFIED & SEALED (SHA-256)
Investigator Finding:	Rapid descent detected near geofence boundary.		

EVID-02: 3D Trajectory Incident Reconstruction (Waypoint #46 at 12:48:30 UTC)



Waypoint Ref:	Waypoint #46	Incident Time UTC:	12:48:30 UTC
Altitude (MSL):	5.2 m	Ground Speed:	0.5 m/s
Flight Heading:	180° (S)	Attitude (Pitch/Roll):	Pitch: +1.2° Roll: +0.3°
GNSS Position:	19.076890°, 72.878400°	Custody Verification:	VERIFIED & SEALED (SHA-256)
Investigator Finding:	Emergency landing touchdown location.		

7. CHRONOLOGICAL MILESTONE TIMELINE

Timestamp (UTC)	Event Type	Mode / State	Forensic Evidence Context
2026-09-05 05:42:27	GPS FIX	N/A	Telemetry fix
2026-09-05 05:42:28	MODE CHANGE	AUTO_MISSION	Telemetry fix
2026-09-05 05:42:28	ARM DISARM	N/A	Telemetry fix
2026-09-05 05:43:37	ARM DISARM	N/A	[health_and_arming_checks] Preflight Fail: VTOL fixed-wing system failure detected
2026-09-05 05:43:37	RTH TRIGGER	N/A	[failsafe] Failsafe activated: entering Hold for 5 seconds
2026-09-05 05:43:37	MODE CHANGE	AUTO_LOITER	Telemetry fix
2026-09-05 05:43:42	RTH TRIGGER	N/A	[failsafe] Failsafe activated
2026-09-05 05:43:42	MODE CHANGE	AUTO_RTL	Telemetry fix
2026-09-05 05:43:42	RTH TRIGGER	N/A	[navigator] RTL: start return at 110 m (69 m above destination)
2026-09-05 05:44:29	MODE CHANGE	POSCTL	Telemetry fix
2026-09-05 05:45:09	ARM DISARM	N/A	[health_and_arming_checks] Preflight Fail: Attitude failure (roll)
2026-09-05 05:45:09	ARM DISARM	N/A	[health_and_arming_checks] Preflight Fail: VTOL fixed-wing system failure detected
2026-09-05 05:45:09	RTH TRIGGER	N/A	[failsafe] Failsafe activated

2026-09-05 05:45:11	ARM DISARM	N/A	[health_and_arming_checks] Preflight Fail: VTOL fixed-wing system failure detected
2026-09-05 05:45:14	ARM DISARM	N/A	[health_and_arming_checks] Preflight Fail: Flight termination active

8. CERTIFICATE OF AUTHENTICITY (LEGAL ADMISSIBILITY)

I hereby certify under Section 63 of the Bharatiya Sakshya Adhiniyam, 2023 (corresponding to Section 65B of the Indian Evidence Act, 1872) and ISO/IEC 27037 standards that the electronic records, telemetry chronologies, and mathematical calculations contained within this report were generated by lawful digital forensic acquisition. The computing systems and software tools used to parse this evidence operated properly throughout the examination with zero alteration to original electronic memory. The cryptographic digests (SHA-256 and BLAKE3) confirm absolute bit-stream integrity of the exhibited evidence.

Examiner Signature: _____

Date: 2026-09-05

Name: Forensic Examiner

Title: Lead Digital Forensics Examiner

Agency: Cyber Forensic Investigation Laboratory (CFSL / State Police)

Official Seal: [AFFIX LABORATORY SEAL]